



Norwegian
University of
Life Sciences

Project Louis

Articulated Robot Arm



Funded by Eik Lab (pretty please)

Authors

Andreas Brustad
Johannes Standal

Eik Lab
Norwegian University of Life Sciences



1 Motivation

Some tasks heavily rely on human intuition and coordination, making them difficult to fully automate. However, these tasks can also be dangerous, and require high precision. While the task can't be fully automated, robotics can provide mitigation of risk factors such as exposure to dangerous chemicals, radiation, electricity and moving parts. The goal of this project is to produce a teleoperated robot arm, protecting humans from hazardous environments.

2 Final vision

Our approach for this project is to create a robot arm mimicking a human hand. It will be remote controlled by haptic gloves akin to lucid gloves, and provide a feedback loop that ensures finger resistance matching the robots grip. This will provide high levels of intuition when dealing with problems outlined in our motivation. For rotational tracking we intend to utilize an IMU. For positional tracking we intend to follow the evolving standards of the VR industry.

3 Learning goals for project

- Learn how to use advanced sensors for data tracking and communication (e-flesh, IMU, etc)
- Learn how to utilize forward and inverse kinematics for a robot arm
- Learn what the industry standards are and how to utilize them (ROS, STM32, etc)
- Learn what tools exist to aid us in the project, and how to utilize them (soldering, 3D-printing, etc)

4 Learning goals for first iteration

For the first iteration we aim to learn how to design functional parts in CAD software and basic STM32 usage. We also aim to test the e-flesh sensor [Pat+25] for automatic adjustment of grip strength. With respect to these rather simplistic goals, and considering our own lack of knowledge, we have decided to make our own simple design rather than following a more advanced recipe. With the basic knowledge gained from this first iteration, we can better judge what the best path forward afterwards is.

5 First iteration / Proof of concept

We will create an articulated robot arm, with the ability to operate in the task space of $\mathbb{R}^3$, focusing on kinematics and remote control. It will be controlled using a simple game controller (xbox, playstation, etc). Featuring methods like forward and inverse kinematics the robot will have multiple control modes. These will be an angular control system and a positional system in our task space related to the robot base coordinate system.



6 Timeline first iteration

Before March:

- Set up software for STM32 programming
- Learn basic STM32 programming
- Set up the ability to control servos with STM32
- Learn basic Fusion skills
- Learn 3D-printing

Before April:

- Have the physical arm ready with servo control and a gripping mechanism
- Have a functional e-flesh sensor on the gripping mechanism

Before May:

- You should be able to control the arm with a game controller

Before June:

- The arm should be able to move the gripper to a specific point based on input coordinates

7 Project links

<https://github.com/JohannesStandal/Eik---Project-Louis>

References

- [Pat+25] Venkatesh Pattabiraman et al. “eFlesh: Highly customizable Magnetic Touch Sensing using Cut-Cell Microstructures”. In: (2025). arXiv: 2506.09994 [cs.R0]. URL: <https://arxiv.org/abs/2506.09994>.